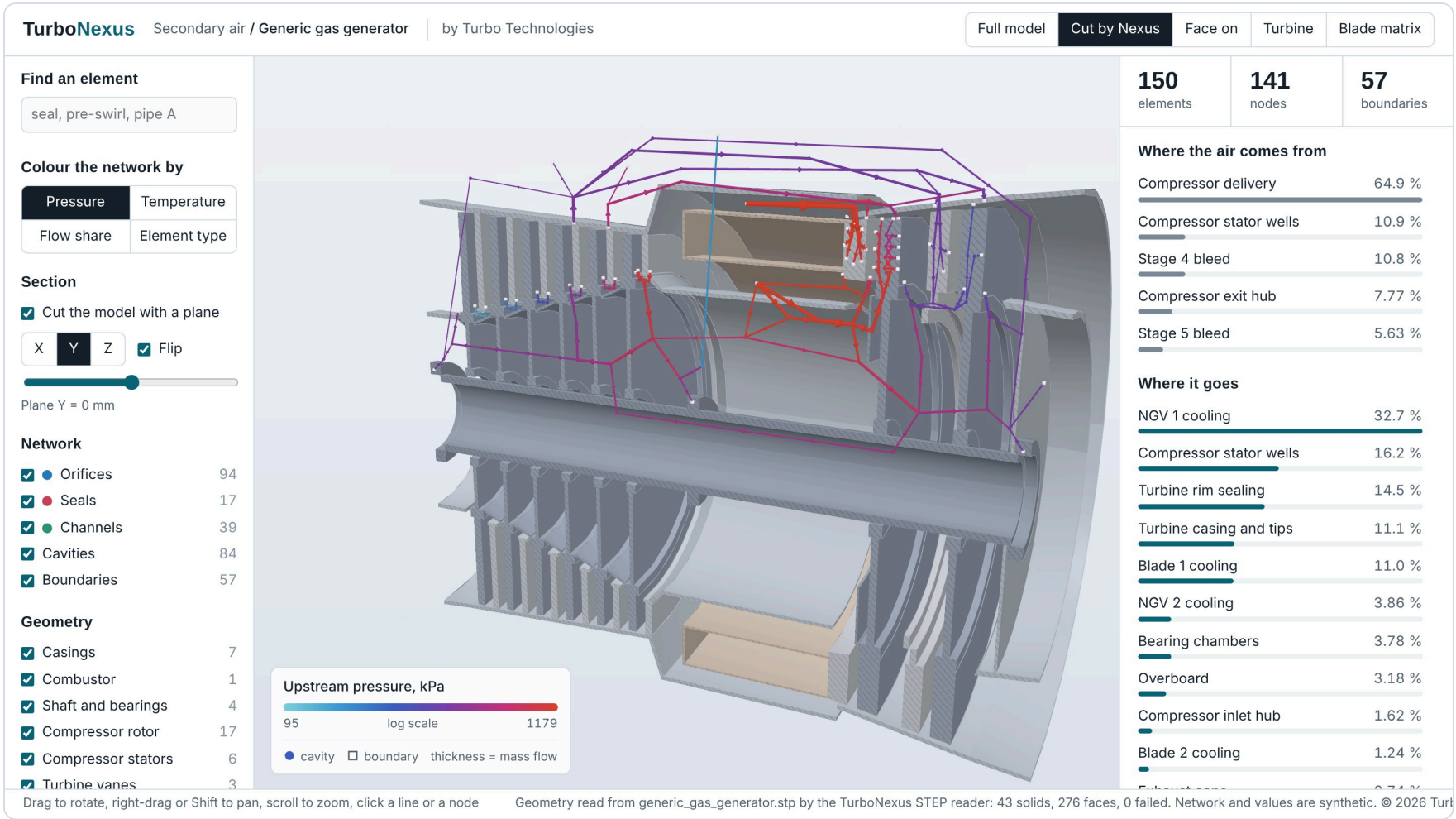
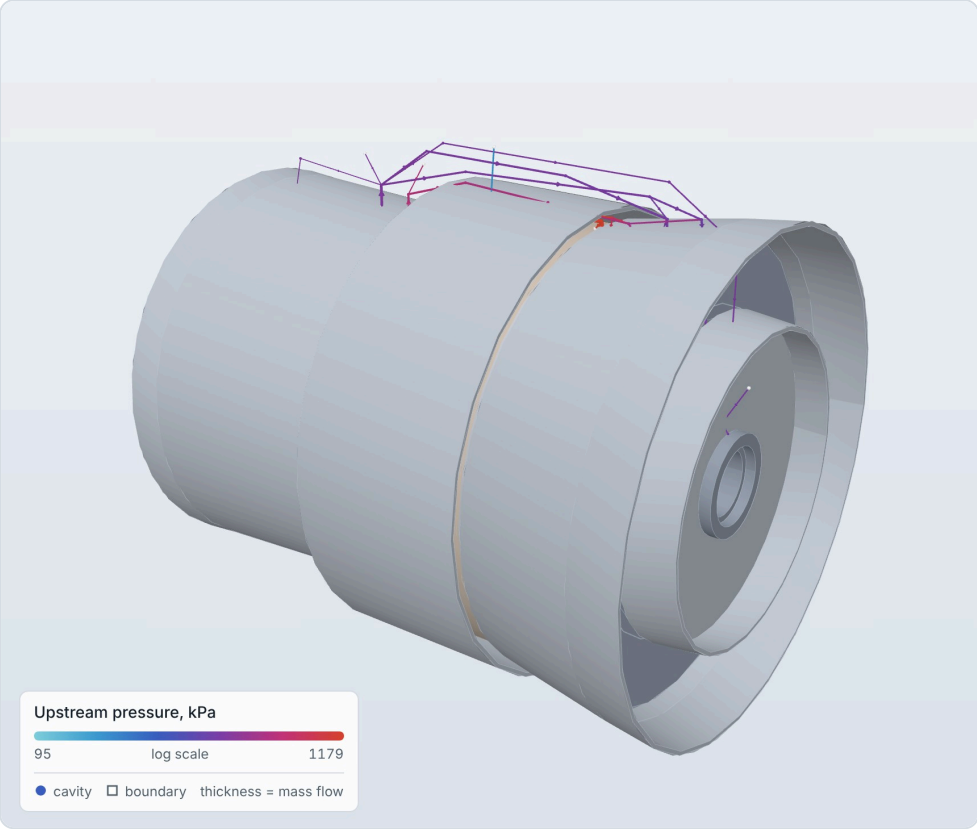
A secondary air network laid on the engine section

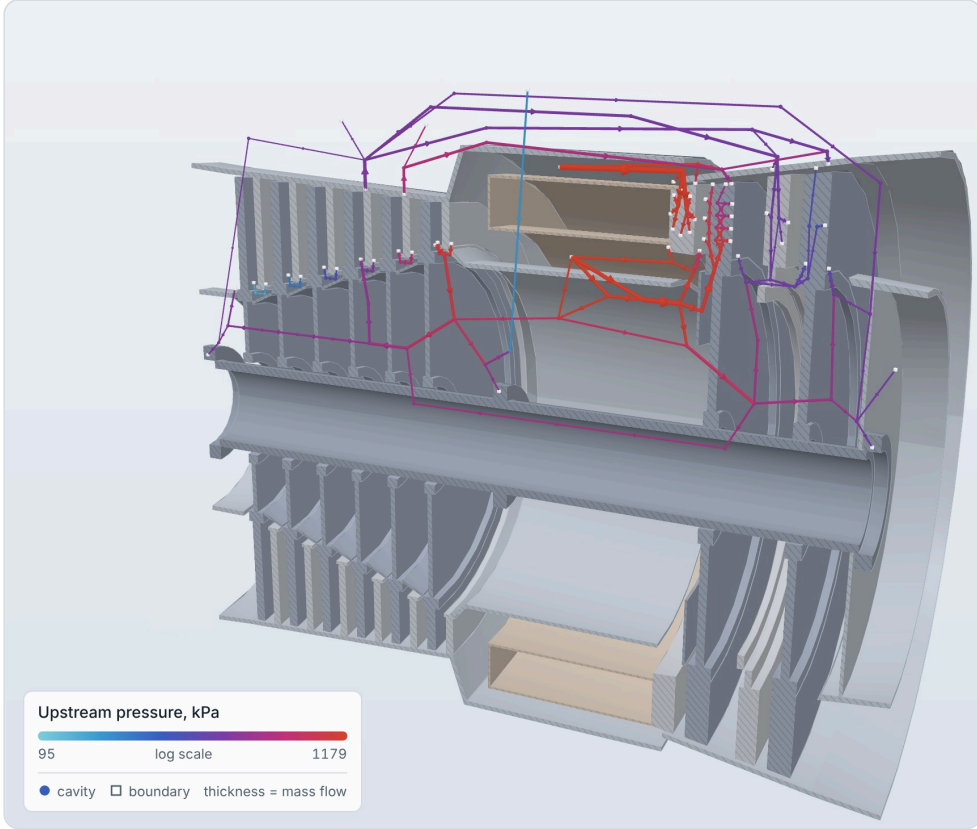
TurboNexus reads the geometry from STEP, cuts it with a plane and draws the 1D air network on the cut: where the air comes from, where it goes and at what pressure.



From a STEP file to a section with the network



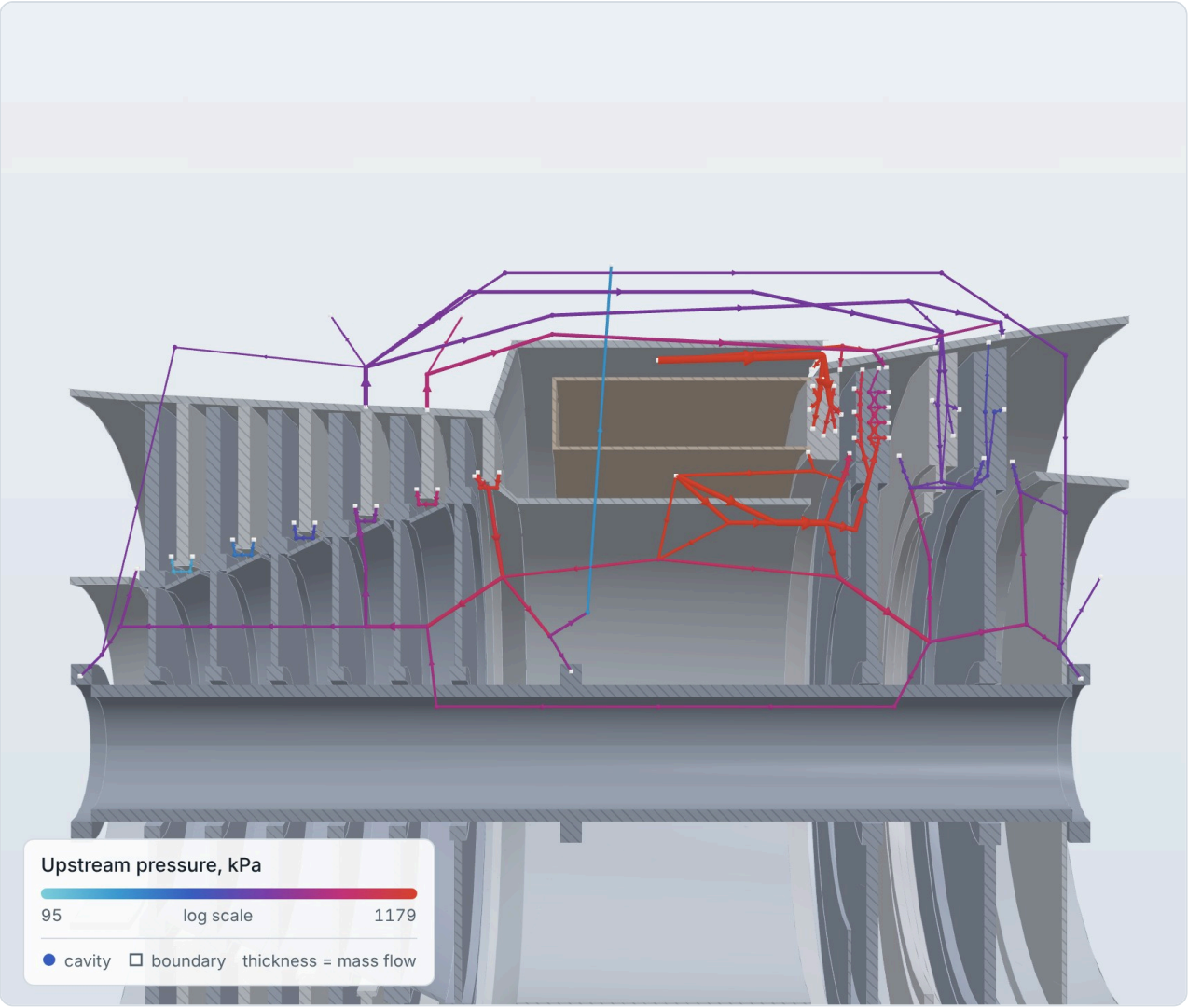
The model as read from STEP: 43 solids in 7 assemblies, 276 faces, 0 failed.



The same model cut by a plane through the axis. Cut faces are hatched; the network lies on the plane.

1. The STEP reader loads the assembly without any CAD system installed.
2. The section tool cuts the model by a plane along X, Y or Z at any position, with filled cut faces.
3. The network of 150 elements (141 nodes) is overlaid on the cut and is never clipped.
4. Line thickness shows mass flow, colour shows pressure, temperature, flow share or element type; arrows show direction.

The air budget at a glance



Section seen face on. Colour is upstream pressure, from 95 to 1179 kPa.

Where the air comes from

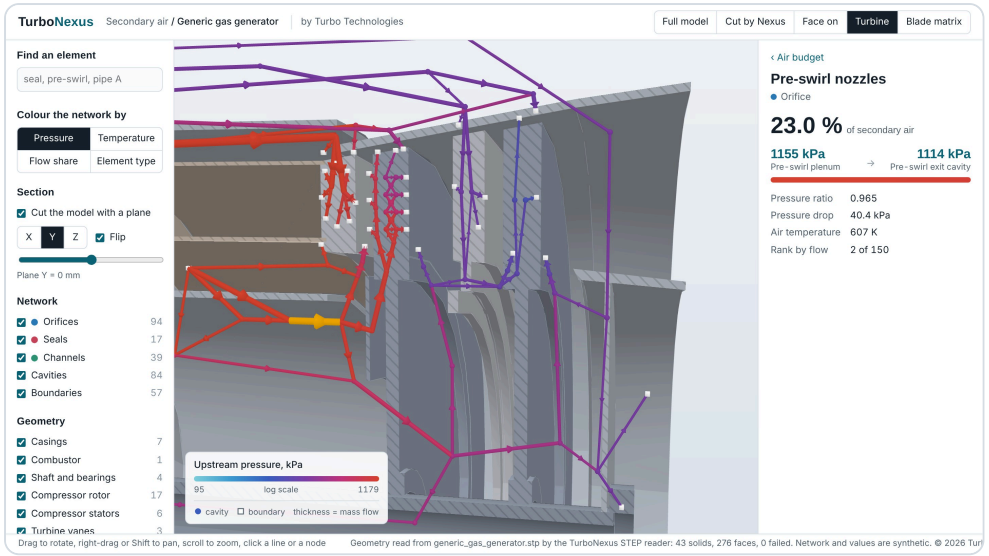
Compressor delivery	64.9 %
Compressor stator wells	10.9 %
Stage 4 bleed	10.8 %
Compressor exit hub	7.77 %
Stage 5 bleed	5.63 %

Where it goes

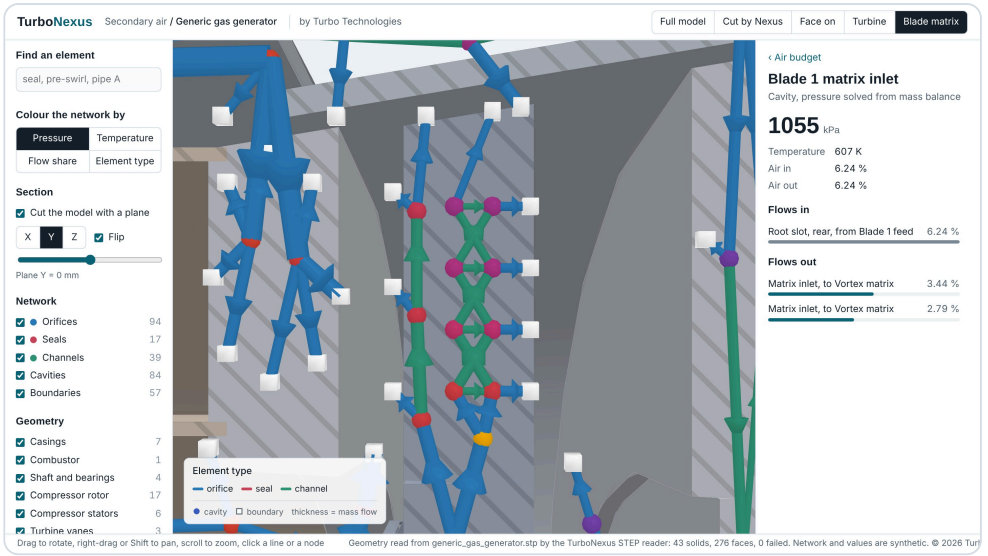
NGV 1 cooling	32.7 %
Compressor stator wells	16.2 %
Turbine rim sealing	14.5 %
Turbine casing and tips	11.1 %
Blade 1 cooling	11.0 %
NGV 2 cooling	3.86 %
Bearing chambers	3.78 %
Overboard	3.18 %
Compressor inlet hub	1.62 %
Blade 2 cooling	1.24 %
Exhaust cone	0.74 %

Shares of all air entering the network. Clicking a row moves the view to those nodes.

Every element and cavity is one click away



An element: its share of the air, pressure before and after with the cavity names, pressure ratio, temperature, rank by flow.



A cavity inside the turbine blade with a vortex matrix: pressure, temperature, flows in and out. Lines coloured by element type.

What is real in this demonstration

- STEP reading, the section with filled cut faces, the overlay and the interface.
- Mass balance of the network: converged, no hot gas enters the cavities.
- Live demonstration: turbo-technologies-s-r-o.github.io/nexus-sas-demo.html

What is not

- The engine: geometry and network are invented, no real design is shown.
- The numbers: a simplified solver with orifice-equivalent elements, no rotation, no heat transfer. They are illustrative.